RV College of Engineering®, Bengaluru – 59
Department of Computer Science and Engineering
DATABASE MANAGEMENT SYSTEMS (CD252IA)

Mid term Report

TITLE: Farmer Resource Sharing System		
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1. Tables created details

1. Farmers table
Purpose:- Stores list of all farmers .
Key Fields - id (PK), clerk_user_id (unique), name, phone

2. Farmer_Groups
Purpose:- Stores list of farmer groups .
Key Fields:- id (PK) , group_name , description

3. Farmer_Group_Members (associative entity)
Purpose:- Stores log of when farmers joined which group .
Key Fields:- id (PK) , farmer_id (FK) , group_id (FK) ,joined_at

4. Land
Purpose :- Stores list of lands and type of land .
Key Fields:- id (PK), farmer_id (FK), group_id (FK), area, location, soil_type

5. Equipment
Purpose:- Stores equipment lending information.
Key Fields:- id (PK), farmer_id (FK), group_id (FK), name ,type, availability

6. Crops
Purpose :- Stores crops type and crop Details
Key Fields :- id (PK), land_id (FK), farmer_id (FK), crop_name,growth_stage

2. Table creation Queries

1. Farmer Table

```
CREATE TABLE IF NOT EXISTS farmers (
    id SERIAL PRIMARY KEY,
    clerk_user_id TEXT UNIQUE NOT NULL,
    name VARCHAR(100) NOT NULL,
    phone VARCHAR(20),
    address TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

2. Farmer_groups Table

```
CREATE TABLE IF NOT EXISTS farmer_groups (
    id SERIAL PRIMARY KEY,
    group_name VARCHAR(100) NOT NULL,
    description TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

3. Farmer_group_members Table

```
CREATE TABLE IF NOT EXISTS farmer_group_members (
    id SERIAL PRIMARY KEY,
    farmer_id INT REFERENCES farmers(id) ON DELETE CASCADE,
    group_id INT REFERENCES farmer_groups(id) ON DELETE CASCADE,
    joined_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

4. Land Table

```
CREATE TABLE IF NOT EXISTS land (
    id SERIAL PRIMARY KEY,
    farmer_id INT REFERENCES farmers(id) ON DELETE SET NULL,
    group_id INT REFERENCES farmer_groups(id) ON DELETE SET NULL,
    area DECIMAL(10,2) NOT NULL,
    location TEXT NOT NULL,
    soil_type VARCHAR(100),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

5. Equipment Table

```
CREATE TABLE IF NOT EXISTS equipment (
    id SERIAL PRIMARY KEY,
    farmer_id INT REFERENCES farmers(id) ON DELETE SET NULL,
    group_id INT REFERENCES farmer_groups(id) ON DELETE SET NULL,
    name VARCHAR(100) NOT NULL,
    type VARCHAR(100),
    availability BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

6. Crops Table

```
CREATE TABLE IF NOT EXISTS crops (
    id SERIAL PRIMARY KEY,
    land_id INT REFERENCES land(id) ON DELETE CASCADE,
    farmer_id INT REFERENCES farmers(id) ON DELETE SET NULL,
    crop_name VARCHAR(100) NOT NULL,
    growth_stage VARCHAR(50),
    expected_yield DECIMAL(10,2),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

3. Table Population Queries

1. Farmers Table

```
INSERT INTO farmers (clerk_user_id, name, phone, address) VALUES
('clerk_001', 'Ramesh Kumar', '9876543210', 'Village Rampur, Karnataka'),
('clerk_002', 'Suresh Patil', '9123456789', 'Taluk Hubli, Karnataka'),
('clerk_003', 'Mahesh Gowda', '9988776655', 'Mandya District, Karnataka'),
('clerk_004', 'Anil Shetty', '9090909090', 'Udupi District, Karnataka');
```

2. Farmer Groups Table

```
INSERT INTO farmer_groups (group_name, description) VALUES
('Organic Farmers', 'Group focusing on organic farming'),
('Rice Growers', 'Farmers specializing in rice cultivation'),
```

('Vegetable Producers', 'Group growing seasonal vegetables'),

('Mixed Crop Group', 'Farmers growing multiple crops');

3. Land Table

INSERT INTO land (farmer_id, group_id, area, location, soil_type) VALUES

(1, 1, 5.50, 'Rampur Field A', 'Black Soil'),

(2, 2, 3.25, 'Hubli East Farm', 'Alluvial Soil'),

(3, 3, 4.00, 'Mandya Plot 3', 'Red Soil'),

(4, 4, 6.75, 'Udupi Coastal Land', 'Laterite Soil');

4. Equipment Table

INSERT INTO equipment (farmer_id, group_id, name, type, availability) VALUES

(1, 1, 'Tractor', 'Heavy Machinery', TRUE),

(2, 2, 'Rice Transplanter', 'Planting Equipment', TRUE),

(3, 3, 'Sprinkler System', 'Irrigation', FALSE),

(4, 4, 'Plough', 'Tillage Tool', TRUE);

5. Crops Table

INSERT INTO crops (land_id, farmer_id, crop_name, growth_stage, expected_yield) VALUES

(1, 1, 'Wheat', 'Vegetative', 20.50),

(2, 2, 'Rice', 'Tillering', 35.00),

(3, 3, 'Tomato', 'Flowering', 15.75);

4. Query execution for data retrieval required for your project

1.Check Farmer Membership in a Group

Explanation:Retrieves records to verify whether a specific farmer belongs to a given farmer group.

```
SELECT * FROM farmer_group_members
    WHERE group_id = ${groupId}
    AND farmer_id = ${farmer_id}
```

2. Fetch Farmers in a Group with Details

Explanation: Returns detailed information of all farmers in a specific group, ordered by their joining date in descending order.

```
SELECT
      fgm.id,
      fgm.group_id,
      f.id AS farmer_id,
      f.name,
      f.phone,
      f.address,
      fgm.joined_at
    FROM farmer_group_members fgm , farmer f
    JOIN farmers f ON f.id = fgm.farmer_id
    WHERE fgm.group_id = ${groupId}
    ORDER BY fgm.joined_at DESC;
```

3.Check Equipment Availability

Explanation:

Fetches the availability status of a specific equipment item using its ID.

SELECT availability FROM equipment WHERE id = ${id}

4. List All Land Records

Explanation:

Retrieves all land records sorted by their creation date, showing the most recently added land first.

SELECT * FROM land ORDER BY created_at DESC